

Inventive Cognition: Analogical Reasoning, Combinatorial Creativity, and Mental Simulation

Executive Summary

Between perception and action lies cognition — the internal process of updating generative models, reasoning about causes, simulating possibilities, and generating candidate solutions. Inventive cognition is not magical inspiration; it is the structured manipulation of generative models through well-characterized cognitive operations. This module examines four principal modes of inventive cognition through the Active Inference lens: analogical reasoning (mapping structural relationships from known to unknown domains), combinatorial creativity (recombining existing model components into novel configurations), systematic inventive thinking rooted in TRIZ methodology (applying cataloged solution patterns to well-defined contradictions), and mental simulation (running the generative model forward to predict consequences of actions not yet taken). Each mode corresponds to a specific operation on the generative model, making inventive cognition systematic, teachable, and improvable.

Learning Objectives

1. Describe analogical reasoning as the transfer of structural relationships between generative models across domains
2. Explain combinatorial creativity as the recombination of generative model components and identify conditions that make combinations productive
3. Analyze TRIZ inventive principles as a curated library of model-transformation operations for resolving contradictions
4. Characterize mental simulation as running the generative model forward in time to evaluate candidate inventions before building them
5. Apply multiple cognitive strategies to a single invention problem and evaluate which strategy produces the most promising results

Key Concepts

1. Analogical Reasoning: Structural Mapping Between Models

Analogical reasoning is perhaps the most powerful cognitive tool available to inventors. It works by identifying structural correspondences between a well-understood source domain and a less-understood target domain, then transferring relational knowledge from source to target. In Active Inference terms, analogical reasoning is the application of a generative model developed in one context to generate predictions in another context.

The structure-mapping theory developed by Dedre Gentner provides the formal framework. An analogy preserves relational structure (how elements interact) while discarding surface features (what the elements look like). When Johannes Gutenberg conceived the printing press, he drew an analogy from the wine press — a device that applied uniform pressure across a surface. The surface features were completely different (grapes vs. ink, barrels vs. paper), but the relational structure was preserved: uniform pressure applied across a surface to transfer a substance from one medium to another.

In Active Inference, analogical reasoning succeeds when two domains share the same high-level generative model structure despite having different low-level parameters. The wine press and the printing press share the same causal graph: force applied to flexible medium against rigid surface produces uniform contact. By identifying this shared structure, Gutenberg could transfer the wine press's engineering solutions (screw mechanism, flat platen, basin containment) to the printing problem.

Effective analogical reasoning requires two cognitive capacities: the ability to abstract away surface features (reducing precision on low-level parameters) and the ability to detect shared relational structure (maintaining precision on high-level causal relationships). Inventors who excel at analogy tend to have broad experience across multiple domains, providing a rich library of source models, combined with the capacity for structural abstraction that allows them to see past surface differences.

The risk of analogical reasoning is false analogy — transferring relational structure that does not actually hold in the target domain. The "horseless carriage" analogy that dominated early automobile design led to cars that looked like carriages, complete with whip sockets and high

ground clearance for avoiding horse manure. The analogy broke down because the relational structure of horse-drawn transportation (low speed, animal-limited power, rough roads) did not map to engine-powered vehicles. Effective inventors test their analogies against evidence, treating them as hypotheses to be validated rather than conclusions to be assumed.

2. Combinatorial Creativity: Recombining Model Components

Arthur Koestler coined the term "bisociation" to describe the creative act of connecting two previously unrelated frames of reference. In Active Inference terms, combinatorial creativity involves taking components from different generative models and assembling them into a new model that generates novel predictions.

The smartphone is the canonical example of combinatorial invention. It combined components from at least six previously separate generative models: telephony (voice communication), computing (data processing), photography (image capture), cartography (spatial navigation), music playback (audio entertainment), and internet browsing (information access). None of these components was individually novel by 2007; the invention was in the combination.

Not all combinations are creative. Random concatenation produces nonsense, not invention. Productive combinations share three characteristics identified by Margaret Boden in her analysis of computational creativity:

Complementarity: The combined components address different aspects of a coherent need. The smartphone components are complementary because they all relate to personal information management while serving different modalities (voice, text, image, location, audio).

Emergence: The combination produces capabilities that none of the components possess individually. A smartphone is not just a phone plus a camera plus a map; it enables entirely new behaviors (posting geotagged photos, navigating with real-time traffic, video calling) that emerge from the interaction of components.

Coherence: The combination can be described by a unified generative model rather than just a list of parts. The smartphone's generative model is "personal computation and communication device" — a single concept that subsumes all components.

In Active Inference, productive combination occurs when components from different models can be integrated into a new model with lower overall free energy than the separate models had individually. The unified model makes better predictions (because it accounts for interactions between components) and requires less overall model complexity (because shared components serve multiple functions).

3. TRIZ and Systematic Inventive Thinking

TRIZ (Teoriya Resheniya Izobretatelskikh Zadach — Theory of Inventive Problem Solving) was developed by Soviet engineer Genrich Altshuller from the analysis of over 200,000 patents. It provides a systematic methodology for inventive cognition based on the observation that most inventive problems share common structural patterns and that solutions to these patterns recur across completely different domains.

The core concept of TRIZ is the "contradiction" — a situation where improving one parameter of a system worsens another. Making an airplane wing stronger (beneficial) makes it heavier (detrimental). Making a knife sharper (beneficial) makes it more fragile (detrimental). In Active Inference terms, a contradiction exists when the generative model predicts that no single action can simultaneously reduce prediction error on two competing variables.

TRIZ's 40 inventive principles are operations that resolve contradictions by transforming the generative model rather than accepting the trade-off. For example:

Principle 1 (Segmentation): Divide the system into parts that can be independently optimized. This transforms the generative model from a monolithic system to a modular one, allowing different modules to satisfy different constraints.

Principle 13 (Inversion): Do the opposite of what is conventionally done. This inverts the generative model's predictions, often revealing that the "obvious" approach is not the only viable one. The Dyson bladeless fan inverts the conventional model of fans (visible rotating blades create airflow) by using hidden impellers and air amplification.

Principle 35 (Parameter change): Change a parameter of the system (temperature, pressure, density, flexibility). This explores the parameter space of the existing generative model, finding operating points that resolve the contradiction.

From an Active Inference perspective, TRIZ is a curated library of model-transformation operations — specific ways to modify a generative model's structure, parameters, or precision weightings to resolve prediction errors that the current model treats as inherent trade-offs. The power of TRIZ is that these operations have been empirically validated across hundreds of thousands of inventive solutions, making them reliable starting points for cognitive exploration.

4. Mental Simulation: Running the Model Forward

Before building a prototype, inventors simulate — they run their generative model forward in time to predict what would happen if a proposed invention existed. This mental simulation serves the same function as physical prototyping (generating predictions that can be compared against desired outcomes) but with vastly lower cost and cycle time.

In Active Inference, mental simulation is the counterfactual evaluation of policies: given a proposed action (building this invention), what sensory outcomes does the generative model predict? This is precisely the mechanism underlying expected free energy evaluation — the process by which agents select actions that are expected to achieve goals and reduce uncertainty.

Nikola Tesla was famous for his ability to conduct detailed mental simulations. He claimed to build and test machines entirely in his imagination, running them mentally for weeks to identify wear patterns and potential failures before constructing a single physical component. While Tesla's accounts may be embellished, the cognitive process he describes — detailed predictive simulation using an internalized physical model — is well-documented in expert inventors.

Mental simulation quality depends directly on generative model accuracy. An inventor whose model accurately represents the relevant physics, user behavior, market dynamics, and manufacturing constraints will generate useful predictions. An inventor whose model is inaccurate will generate plausible but misleading predictions — the cognitive equivalent of a beautiful but wrong theory.

This creates an important feedback loop between mental simulation and physical prototyping. Mental simulation identifies the most promising candidate designs, reducing the number of

physical prototypes needed. Physical prototyping tests the mental simulation's predictions, updating the generative model and improving future mental simulations. The most effective inventors rapidly alternate between mental and physical simulation, using each to calibrate the other.

5. Cognitive Fluency and the Generation of Alternatives

Inventive cognition requires not just the ability to generate one solution but the ability to generate many — to explore the space of possibilities before converging on a single design. This capacity for divergent thinking, or cognitive fluency, is essential because the first solution an inventor conceives is rarely the best.

In Active Inference terms, cognitive fluency reflects the agent's willingness to maintain high uncertainty (high entropy in the policy distribution) during the generative phase. Rather than immediately committing to the policy with the lowest expected free energy, the fluent thinker entertains multiple policies simultaneously, exploring a broader region of the solution space before converging.

The SCAMPER method (Substitute, Combine, Adapt, Modify, Put to other uses, Eliminate, Reverse) provides a systematic scaffold for cognitive fluency. Each SCAMPER operation is a specific transformation of the generative model that produces a variant solution. By applying all seven operations to a starting concept, the inventor generates at minimum seven alternative designs, each exploring a different dimension of the solution space.

Research in design cognition consistently shows that inventors who generate more alternatives in the early stages of a project produce better final designs. This is not because quantity produces quality directly, but because breadth of exploration increases the probability of encountering a region of the solution space where a genuinely superior solution exists. The cognitive discipline of delaying convergence — continuing to generate alternatives even when a satisfactory solution has been found — is one of the most difficult but most rewarding practices in inventive work.

Applications

Case Study 1: Velcro and the Power of Analogical Cognition

George de Mestral's invention of Velcro (1941-1955) is often told as a story of accidental perception, but the cognitive work that followed the initial observation was equally important and far more deliberate.

After perceiving the hook-and-loop mechanism of burrs, de Mestral had to perform several cognitive operations. First, he abstracted the structural principle: tiny hooks engage with loops to create a detachable bond. This abstraction required stripping away the biological context (plant reproduction, seed dispersal) to isolate the mechanical principle (reversible attachment via hook-loop engagement).

Second, he performed an analogical mapping from the biological domain to the manufacturing domain. Hooks could be manufactured as tiny curved filaments; loops could be manufactured as fabric. But the analogy was imperfect — biological hooks are grown in three dimensions with complex geometry, while manufacturing processes at the time could only produce regular, repeating patterns. This mismatch required cognitive problem-solving: how to manufacture hooks that were sufficiently irregular to engage with loops reliably but sufficiently regular to be mass-produced.

Third, he mentally simulated use cases: clothing fasteners, shoe closures, industrial attachments. Each use case placed different demands on the hook-loop mechanism (holding force, cycle durability, flexibility), requiring different model parameters. The mental simulation identified that nylon was the most promising material — a prediction that required extensive physical prototyping to validate.

The full invention took 14 years from initial observation to commercial product, most of that time spent on cognitive and engineering work that transformed a biological observation into a manufactured product. The creative perception was necessary but not sufficient; it was the cognitive processing — analogy, abstraction, mental simulation, and systematic problem-solving — that turned observation into invention.

Case Study 2: TRIZ in Action — Samsung's Display Innovation

Samsung's development of flexible OLED displays provides a modern example of TRIZ-style systematic inventive thinking. The core contradiction was: display screens should be large (for viewing quality) AND portable (for carrying convenience). Increasing size worsens portability; improving portability requires reducing size. This is a classic TRIZ physical contradiction.

TRIZ suggests several principles for resolving physical contradictions. Samsung's engineers applied **Principle 15 (Dynamism)**: make a rigid object flexible so it can change configuration. Rather than choosing between large and portable, they created a display that could be both — large when unfolded for viewing, small when folded for carrying.

This resolution required extensive mental simulation: How would repeated folding affect pixel integrity? What materials could survive millions of fold cycles? How would the user interface adapt to different form factors? Each simulation generated predictions that guided material science research (flexible substrates, foldable encapsulants) and user experience design (continuity between folded and unfolded states).

The TRIZ framework did not provide the specific engineering solution (that required thousands of hours of materials research and prototyping). What it provided was the cognitive direction: look for a way to make the display dynamically reconfigurable rather than accepting the size-portability trade-off. This cognitive reframing from "choose between large and portable" to "make a display that is both by being flexible" was the inventive cognitive act that TRIZ facilitated.

Cross-References

- **Module 03 (Creative Perception)**: Perception provides the raw material for cognition; cross-domain pattern recognition (Module 03) feeds analogical reasoning (this module)
- **Module 05 (Creative Action)**: Cognition generates candidate solutions; action (prototyping) tests them, creating the cognition-action loop
- **Module 02 (Creative Agent)**: Different inventor archetypes favor different cognitive strategies; the Methodical Engineer favors TRIZ, the Tinkerer favors combinatorial play

- **Module 08 (Planning):** Mental simulation is the cognitive basis for planning; this module describes the mechanism, Module 08 describes the strategy
- **Section 1, Module 04 (Cognition in Invention):** Foundational cognitive concepts; this module extends them with specific inventive cognitive strategies

Summary Table

Concept	Active Inference Term	Invention Application	Key Insight
Analogical reasoning	Generative model transfer across domains	Gutenberg: wine press to printing press	Shared causal structure enables cross-domain solution transfer
Combinatorial creativity	Recombination of model components	Smartphone: telephony + computing + camera + GPS	Productive combinations create emergent capabilities
TRIZ principles	Cataloged model-transformation operations	Samsung: rigid display to flexible display via Dynamism	Contradictions are resolved by transforming the model, not accepting trade-offs
Mental simulation	Counterfactual policy evaluation	Tesla: testing machines in imagination before building	Model accuracy determines simulation value
Cognitive fluency	High-entropy policy distribution	SCAMPER: generating multiple solution variants	More alternatives explored = better final design
Contradiction resolution	Free energy reduction across competing variables	Size vs. portability, strength vs. weight	Inventive cognition reframes trade-offs as transformation opportunities

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